G3. RISK MANAGEMENT PLAN

Version 02

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1. Introduction

This document is the risk management plan for the VR-CARDIO family (that includes Ref: VR-CARDIO-01 and Ref: VR-CARDIO-02) to define the methodology for identifying, controlling, reducing and, if possible, eliminating potential hazards resulting from process and/or product failure.

Define the methodology to ensure that risks and opportunities are identified and managed.

These analyses must be updated when device risks and technologies change. The main methods and procedures selected are focused on ISO 14971:2019 and ISO/TR 24971:2020.

2. Acronyms and definitions

Risk: combination of the probability of occurrence of harm and the severity of that harm.

Risk management: systematic application of management policies, procedures and practices to the tasks of analysing, evaluating, controlling and monitoring risk.

Injury: physical lesion, health damages, property or environmental damages.

Harm: Health damage, including the damage that can occur from loss of product quality or availability.

Hazard: The potential source of harm.

Hazardous Situation: circumstance in which people, property or the environment are exposed to one or more hazards (ISO 14971)

Adverse event: product defect (i.e. malfunction or failure, deterioration in characteristics or performance, or inadequacy of labelling or of instructions for use) that, directly or indirectly, has led or might have led to serious consequences, namely deterioration in the state of health of the patient, user or another person. Also called an incident.

Severity: A measure of the possible consequences of a hazard.

Probability of Occurance: Rank the probability of a failure occurring during the conception, manufacturing and commercialization of the product

Risk Assessment: Set of operations that allows the evaluation of the potential harm (damage) or injury.

Risk Analysis: Study that aims to determine the possible risks that may occur during the operational phase of a product.

Risk Control: Application of a set of operations that allows evaluation, control and monitoring of risks associated with a particular product or process in order to ensure that the identified risks do not occur or has a very low probability.

Residual Risk: Risk remaining after risk control measures have been taken.

Risk Management Team (RTM): Representatives of several functions that contribute to risk evaluation according to their specific knowledge.

RMT: Risk Management Team

RI: Risk Index

S: Severity

P: Probability of Occurrence

RMP: Risk Management Plan

RMR: Risk Management Report

3. Responsibilities

Department	Responsibilities
Executive Director	Assess potential risks from the perspective of usage and customer complaints, such as feedback.
Quality guarantee	Evaluate possible risks from the point of view of the clinical area and compliance with the requirements defined in the regulations and laws in force for the product.
IT & Robotics	Assess potential risks from a hardware implementation perspective. Evaluate potential risks from a software implementation perspective. Assess possible risks from a production point of view and “control procedures for non-conforming products” and implementation of the “corrective and preventive action control strategy” in the production process.
Investigation and development	Assess potential risks from a hardware and software implementation perspective.

	Assist in the creation, review and approval of risk management activities.
Sales	Responsible for providing information, as needed, regarding marketing, usage, customer feedback, and new risks.
Healthcare professionals	Assess potential risks from the perspective of the clinical environment and clinical practices and provide this information to the manufacturer.

4. Assignment of activities

Risk management activities	Frequency or trigger	Responsible (Participants)	Notes
Maintain the risk management plan	Annually	Quality Manager	Review of the Risk Management Plan
Prepare and maintain the Risk Management Matrix	Annually	Risk Management Team	Reactive updates from change control, complaints and reportable events, receiving inspection, production, CAPA and other applicable processes
Collect and analyze production and post-production information	Annually	Quality Director and Compliance Manager	In accordance with: - Information provided from the production activity. - Post-market surveillance plan
Internal audit of the risk management process	Annually	Auditor	--------

Risk Management Review	Annually	Address	--------

5. Review of Risk Management activities

Risk identification begins by identifying the functional aspects of the device. From this, information about possible problems is collected and a list of potential harms is drawn up. However, this list cannot be considered exhaustive, as there are risks that are not obvious and can only be detected through subsequent use experience with a wide range of users.

It is crucial to regularly review and update the risk assessment. This review must be carried out at least once a year and recorded in the risk matrix. Additionally, it should be updated when necessary, such as in the event of changes to the medical device or when post-sales surveillance data indicates the need for re-evaluation.

The risk analysis should be based on the experience of operators and feedback from customers and users. Information for risk assessment can be obtained from various sources, such as customer complaints, satisfaction surveys, customer and user opinions on product performance, identified improvement opportunities, results of post-sales surveillance and clinical follow-up, technical updates and regulations, expert opinions, usability tests, published standards and incident reports available.

6. Risk Management

The risk management process is based in ISO 14971 - Medical devices — Application of risk management to medical devices and in Medical Device Regulation (EU) 2017/745. The risk analysis is performed using the technique FMEA - Failure Mode and Effect Analysis, according to the mentioned in annex B of ISO/TR 24971 - Medical devices — Guidance on the application of ISO 14971 (as described in PNT-002 Risk Management Procedure).

6.1. Risk assessment

For this evaluation, all possible effects of the risks will be identified and will be used to evaluate each of them's Probability and Severity in each case. The RPN (Risk Priority Number) has been calculated for each type of failure/danger.

The risk analysis will be developed as follows:

- Description of possible risks

- Causes and effects on the product and/or patient and/or end user
- Risk Priority Number: according to the following parameters:
    - Probability (P): expected frequency of the identified risk occurring.

Probability of the harm	Description	Score
Improbable	≥ 1 in 1 000000 ($<10^{-6}$)	1
Remote	≥ 1 in 100000 ($<10^{-5}$ and $\geq 10^{-6}$)	2
Occasional	≥ 1 in 10000 ($<10^{-4}$ and $\geq 10^{-5}$)	3
Probable	≥ 1 in 1000 ($<10^{-3}$ and $\geq 10^{-4}$)	4
Frequent	≥ 1 in 100 ($\geq 10^{-3}$)	5

    - Severity (S): severity of the identified risk occurring, assesses the level of consequences.

Severity of the harm	Description	Score
Negligible	Results in inconvenience or temporary discomfort	1
Minor	Results in temporary injury or impairment not requiring medical or surgical intervention	2
Serious / Major	Results in injury or impairment requiring medical or surgical intervention	3
Critical	Results in permanent impairment or irreversible injury	4
Catastrophic / Fatal	Results in death	5

6.2. Risk acceptance

Once the evaluation of each risk has been made, the RPN (Risk Priority Number) is obtained by applying the following formula:

$$\mathbf{RPN = P \times S}$$

Each risk estimated must be systematically evaluated, considering that risks should be reduced as far as possible (AFAP) without adversely affecting the benefit-risk ratio. Therefore, even if the risk is acceptable is necessary to implement measures to bring the risk to its lowest level.

The acceptability of the resulting RPN is based on pre-established cut-off values, such as:

RPN <4	Acceptable risk
RPN ≥ 4 but < 15	Tolerable risk
RPN ≥ 15	Unacceptable risk

Note:

Acceptable Risks - Additional controls can be considered, but are not required.

Tolerable Risks – Risks should be reduced as far as possible (AFAP).

Unacceptable Risks - Design changes are needed to mitigate risk, or a benefit-risk analysis is needed to further assess benefit-risk.

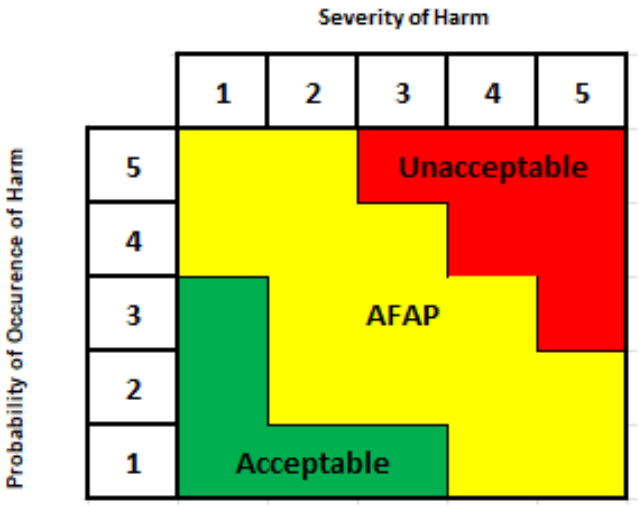

Image 1 – Risk evaluation matrix

6.3. Review of the potential hazard plan

The Risk Analysis is reviewed at least annually, taking into account the post-production experience gained. The reasons leading to exceptional revisions of the Risk Analysis are at least the following:

- Non-effective control measures: occurrence of danger;
- Change in the process of product realization: improvements resulting from the previous point, technological changes;
- Product change;
- Regulatory and state-of-the-art change;

- New products.
- After this step, a new version of the Risk Management Report should be issued.

6.4. Verification activities

Verification activities will be carried out to verify whether the risk control has been effectively implemented, and to ensure that no other risks have been introduced and that the actions implemented actually produce a reduction in risk, and whether that reduction is at its lowest level. The verification activity will be based on the collection of clinical data and/or studies carried out on other products.

6.5. Production and post-production information

Information related to the production phase will be collected during the development process and in process control, where risks related to software development can be identified.

The collection of information related to the device in the post-production phase aims to obtain information related to the use, safety of the product and other relevant aspects. This information can be obtained from different sources such as users, complaints, incident reports, etc. The information collected is analyzed and processed and may result in a change to the list of harms initially identified.

6.6. Usability

The usability of VR-CARDIO family will be made according to IEC 62366 – Medical devices: Application of usability engineering to medical devices series.

This process starts with the product conception and finishes in the end of the product use, passing through the different phases of development, labelling, distribution, and use.

It's focused on the product security and try to understand the professional and the factors that can have some influence in the risk increase.

In this scope, the following parameters must be analyzed:

- Frequent function of the product.
- Identification of harms related to the intended use and clinical purpose.
- Identification of harms related to the intended users
- Identification of harms related to usability.

- Identification of characteristics related to the security of the device during the use.
- Identification of known harms or predictable.

6.7. Overall Residual Risk

If a residual risk does not meet established acceptability criteria, the available data are examined to determine whether the benefits of the intended use outweigh the residual risk. However, it is also essential to ensure that risks are assessed based on whether they have been reduced to its lowest possible level without negatively affecting the benefit-risk ratio.

Assessing residual risk may involve collecting and reviewing data on similar medical devices available on the market. If the residual risk is considered acceptable, users should be informed of significant residual risks.

6.8. Results presentation and data review

The risk analysis is reviewed at least once a year or whenever:

I. there is the intention to introduce changes in the device, product or process, either by internal initiative or in response to market information

II. post-marketing surveillance data, including user feedback, reveals new risk, harm or hazard

III. changes in the state of the art